

MINGRUI ZHANG

MTS
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EDUCATION

University of Washington, Seattle

Sep. 2017 - June 2022

Ph.D. in Information Science, The Information School

Focus: Human-Computer Interaction

Advisor: Jacob O. Wobbrock

Tsinghua University, Beijing

Aug. 2013 - Jul. 2017

B.Eng. with Honors, Department of Computer Science and Technology

Minor in journalism

The Chinese University of Hong Kong, Hong Kong

Sep. 2015 - Dec. 2015

Exchange student, Department of Computer Science and Engineering

RESEARCH INTEREST

Large Language Models, Agentic AI and Human-Computer Interaction

My research interests center on Agentic AI and Large Language Models, focusing on 1) building LLM-based agents capable of reasoning, planning, and interacting with complex environments — search, tools, terminals — much like humans do; 2) advancing AI-mediated interaction paradigms that make intelligent systems more intuitive, communicative, and human-like companions.

PROFESSIONAL EXPERIENCE

Reflection.ai. New York, NY

Nov. 2025 - Present

Member of Technical Staff

- Built and owned the mid-training data curation pipeline with GPU clusters for asynchronous inference, transforming in-house and open-source raw data into 2M+ high-quality data pairs via automated rewriting and synthetic data generation.
- Led data curation strategy for math/STEM reasoning, designing a task-difficulty identification framework and iterative curation-training loop — surpassing Qwen3-14B on math/STEM benchmarks including AIME.
- Founded 3 internal ML tools (trace debugging, eval comparison, data inspection/visualization) adopted as core training infrastructure across the 80+ person modeling org; built agentic workflows for ML observability and data onboarding monitoring.
- Refactored internal evaluation frameworks to support third-party agentic LLM evaluation, cutting eval onboarding time from weeks to days.

Meta Platforms Inc. New York, NY

Oct. 2024 - Nov. 2025

Modern Recommendation Systems of Meta AI

Research Scientist

- User understanding, cohort modeling, and interest embedding for Instagram and Facebook.
- Model architecture design for LLM based contextual embedding for user biography, integrated with Meta's HSTU generative recommendation model.
- Post-training (SFT + RL) of ranking LLM agents, achieving 1st rank on TableBench benchmark for table understanding with our RankAgent model.

Meta Reality Labs. New York, NY

Oct. 2022 - Oct. 2024

Input Exploration Team

Research Scientist

- Research lead for text input interaction on AR glasses (Meta Rayban Display) with gestural interface based on EMG signals, including keyboard, conversational dictation, and EMG handwriting.
- Input interaction prototyping with hand + gaze for next generation mix reality headset (Quest).

Apple Inc. Seattle, WA

Jun. 2021 - Sep. 2021

Advised by Huy Viet Le and Tim Paek

Research Intern

- Gaze-based dictation interaction look to dictate/edit for Apple Vision Pro, including interface prototype and user intent classification modeling.

Facebook Reality Labs. New York, NY

Jun. 2020 - Oct. 2020

Advised by Adam Berenzweig

Research Intern

- EMG-based QWERTY-style text entry interactions, enabling ten-finger typing on any surface.
- Enhanced neural network structure and data collection pipeline with CTC loss for gesture classification.

Google Inc. Mountain View, CA

Oct. 2019 - Dec. 2019

Advised by Shumin Zhai

Research Intern

- Sentence level Gboard auto-correction features, see publication C.3.

Momenta.ai Inc. Beijing

Dec. 2016 - Jun. 2017

Advised by Ji Liang

Research Intern

- Enhanced road segmentation models, increasing recall and accuracy by 5%.
- Optimized framework speed from 8fps to 100fps.

MailTime Inc. Beijing

Feb. 2016 - Jun. 2016

iOS Intern

- Designed and implemented the iOS interaction logic and UI, improving user retention by 25%.

Chestnut Tech Inc. Beijing

Oct. 2014 - Jan. 2016

Co-founder, iOS Developer

- Developed “Parocam” application on iOS, focusing on face transform algorithms and UI design.

SELECTED PEER-REVIEWED CONFERENCE PAPERS

For the full publication list, go to my google scholar page

- C.5 Patrick Kaifosh, Thomas R. Reardon & CTRL-labs at Reality Labs (including **Mingrui Zhang**) (2025) A generic non-invasive neuromotor interface for human-computer interaction. *Nature*.
- C.4 **Mingrui “Ray” Zhang**, Shumin Zhai, Jacob O. Wobbrock. (2022) TypeAnywhere: A QWERTY-Based Text Entry Solution for Ubiquitous Computing. *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI '22)*. New Orleans, USA (April 30 - May 6, 2022). New York: ACM Press. ^[25%]
- C.3 **Mingrui “Ray” Zhang**, Shumin Zhai. (2021). PhraseFlow: Designs and Empirical Studies of Phrase-Level Input. *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI '21)*. Yokohama, Japan (May 8-13, 2021). New York: ACM Press. ^[26%]
- C.2 **Mingrui “Ray” Zhang**, He Wen, Jacob O. Wobbrock. (2019). Type, Then Correct: Intelligent Text Correction Techniques for Mobile Text Entry Using Neural Networks. *Proceedings of the 32nd Annual ACM Symposium on User Interface Software & Technology (UIST '19)*. New York: ACM Press. ^[24%]
- C.1 Xin Yi, Chun Yu, **Mingrui “Ray” Zhang**, Sida Gao, Ke Sun, Yuanchun Shi. (2015). ATK: Enabling Ten-Finger Freehand Typing in Air Based on 3D Hand Tracking Data. *Proceedings of*

the 28th Annual ACM Symposium on User Interface Software & Technology (UIST '15). New York: ACM Press. ^[24%]

HONORS & AWARDS

Ford Fellowship, University of Washington	2022
Best Paper Award, ACM CHI	2019
Excellent Graduate of the CST Department, Tsinghua	2017
The National Scholarship of China (Top 1%)	2016
1st Winner of the National Database Conference Cup	2016